

## JASPER WARDEN, AB, Canada

WMO: 714860

Lat: 52.9267N

Lon: 118.0297W

Elev: 3346

StdP: 13.00

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -24.0      | -16.2 | -31.3                         | 1.1 | -23.1 | -24.0 | 1.6 | -14.8 | 14.2                  | 5.1  | 12.1 | 8.0  | 2.8                   | 190  | 0.625 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 27.8                   | 85.0            | 58.1 | 80.7 | 56.9 | 76.8 | 55.5 | 60.0                | 78.2 | 58.4 | 75.8 | 56.9 | 72.8 | 5.4                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 54.7                            | 72.2 | 60.8 | 52.7 | 67.0 | 59.8 | 50.9 | 62.6 | 59.0 | 27.8          | 78.4 | 26.7 | 75.8 | 25.7 | 73.1 | 67.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|
| 1%                | 2.5% | 5%  | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |
|                   |      |     | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |
| 13.1              | 11.0 | 9.3 |                            | -32.5 | 90.6               | 6.9 | 3.2                                                | -37.4 | 92.9       | -41.5 | 94.8       | -45.4 | 96.6       | -50.4 | 98.9 |
|                   |      |     | DB                         | -32.8 | 62.9               | 6.7 | 2.1                                                | -37.7 | 64.4       | -41.6 | 65.6       | -45.4 | 66.8       | -50.3 | 68.3 |
|                   |      |     | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 37.8   | 16.5  | 20.3  | 28.7  | 38.1  | 47.1  | 54.2  | 59.5  | 57.9  | 49.9  | 38.9  | 26.2  | 15.8  |
|                                                                        | DBStd       | 18.18  | 15.13 | 13.12 | 11.14 | 6.79  | 6.59  | 5.17  | 5.53  | 5.51  | 6.84  | 7.56  | 11.33 | 13.61 |
|                                                                        | HDD50       | 5265   | 1039  | 833   | 662   | 359   | 135   | 16    | 4     | 5     | 84    | 351   | 715   | 1062  |
|                                                                        | HDD65       | 9938   | 1504  | 1253  | 1127  | 807   | 556   | 325   | 185   | 226   | 453   | 810   | 1165  | 1527  |
|                                                                        | CDD50       | 824    | 0     | 0     | 0     | 2     | 44    | 143   | 298   | 250   | 82    | 5     | 0     | 0     |
|                                                                        | CDD65       | 23     | 0     | 0     | 0     | 0     | 0     | 2     | 14    | 6     | 1     | 0     | 0     | 0     |
|                                                                        | CDH74       | 1450   | 0     | 0     | 0     | 1     | 44    | 157   | 643   | 483   | 120   | 2     | 0     | 0     |
|                                                                        | CDH80       | 408    | 0     | 0     | 0     | 0     | 2     | 34    | 214   | 135   | 23    | 0     | 0     | 0     |
|                                                                        | WSAvg       | 3.6    | 3.5   | 3.7   | 4.0   | 4.1   | 4.2   | 3.7   | 3.5   | 3.4   | 3.5   | 3.4   | 3.4   | 3.2   |
|                                                                        | PrecAvg     | 16.3   | 1.3   | 0.7   | 0.7   | 0.9   | 1.3   | 2.1   | 2.2   | 2.1   | 1.6   | 1.3   | 1.1   | 1.1   |
| (15) Precipitation                                                     | PrecMax     | 21.2   | 3.9   | 2.4   | 1.9   | 1.9   | 3.8   | 4.3   | 5.4   | 5.1   | 4.3   | 2.8   | 2.4   | 3.4   |
|                                                                        | PrecMin     | 12.9   | 0.1   | 0.0   | 0.1   | 0.1   | 0.4   | 0.7   | 0.3   | 0.7   | 0.6   | 0.3   | 0.2   | 0.2   |
|                                                                        | PrecStd     | 2.4    | 0.9   | 0.6   | 0.4   | 0.5   | 0.7   | 1.0   | 1.3   | 1.1   | 0.7   | 0.6   | 0.6   | 0.7   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 44.3  | 47.8  | 58.0  | 70.0  | 78.9  | 84.5  | 90.9  | 88.8  | 83.5  | 68.9  | 52.9  | 42.9  |
|                                                                        |             | MCWB   | 36.6  | 37.0  | 42.1  | 49.0  | 52.9  | 56.9  | 59.6  | 59.8  | 56.1  | 50.0  | 43.2  | 36.1  |
|                                                                        | 2%          | DB     | 40.6  | 43.7  | 51.1  | 62.8  | 74.0  | 78.3  | 85.8  | 83.9  | 77.5  | 61.2  | 46.3  | 38.6  |
|                                                                        |             | MCWB   | 34.7  | 35.0  | 38.8  | 45.6  | 51.3  | 55.1  | 58.4  | 58.3  | 55.3  | 47.0  | 39.7  | 33.8  |
|                                                                        | 5%          | DB     | 37.8  | 40.2  | 47.0  | 57.5  | 68.9  | 73.9  | 81.5  | 79.6  | 71.7  | 55.5  | 42.5  | 35.6  |
|                                                                        |             | MCWB   | 33.1  | 32.8  | 36.9  | 42.9  | 49.8  | 53.6  | 57.3  | 57.1  | 53.5  | 44.9  | 37.4  | 31.8  |
|                                                                        | 10%         | DB     | 34.8  | 36.9  | 43.7  | 52.9  | 64.4  | 69.5  | 76.6  | 75.1  | 65.0  | 51.0  | 39.3  | 32.9  |
|                                                                        |             | MCWB   | 31.1  | 31.2  | 34.8  | 40.3  | 48.1  | 51.9  | 56.2  | 55.9  | 50.7  | 42.6  | 35.0  | 29.8  |
|                                                                        | 0.4%        | WB     | 37.9  | 38.6  | 43.2  | 49.4  | 54.8  | 59.0  | 62.1  | 62.3  | 58.1  | 51.7  | 44.3  | 37.0  |
|                                                                        |             | MCDB   | 42.2  | 44.8  | 55.1  | 67.5  | 72.6  | 77.8  | 80.3  | 79.2  | 76.5  | 65.1  | 50.6  | 40.5  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 35.5  | 35.8  | 40.0  | 46.5  | 52.7  | 56.8  | 60.2  | 60.2  | 56.0  | 48.9  | 40.8  | 34.4  |
|                                                                        |             | MCDB   | 39.4  | 42.1  | 48.6  | 60.3  | 69.8  | 72.7  | 78.7  | 78.1  | 74.0  | 58.6  | 45.6  | 37.8  |
|                                                                        | 5%          | WB     | 33.5  | 33.6  | 37.7  | 43.7  | 50.9  | 55.1  | 58.7  | 58.5  | 54.0  | 45.9  | 38.1  | 32.3  |
|                                                                        |             | MCDB   | 37.2  | 39.0  | 45.6  | 55.5  | 66.8  | 69.6  | 76.6  | 75.5  | 69.2  | 53.9  | 41.6  | 35.3  |
|                                                                        | 10%         | WB     | 31.3  | 31.5  | 35.6  | 41.2  | 49.0  | 53.3  | 57.2  | 56.8  | 51.8  | 43.5  | 35.5  | 29.8  |
|                                                                        |             | MCDB   | 34.6  | 36.5  | 42.5  | 51.6  | 62.6  | 66.3  | 73.3  | 72.1  | 64.0  | 50.2  | 38.6  | 32.6  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 19.2  | 22.5  | 21.6  | 24.5  | 26.5  | 25.2  | 27.8  | 29.0  | 25.9  | 20.9  | 16.1  | 17.9  |
|                                                                        |             | MCDBR  | 17.2  | 23.2  | 25.8  | 34.3  | 38.4  | 36.6  | 40.7  | 40.5  | 40.4  | 29.8  | 17.9  | 15.6  |
|                                                                        |             | MCWBR  | 11.9  | 15.1  | 14.9  | 18.6  | 18.5  | 15.9  | 16.1  | 17.2  | 19.2  | 16.8  | 11.8  | 11.5  |
|                                                                        | 5% WB       | MCDBR  | 16.1  | 21.6  | 23.8  | 31.4  | 34.7  | 30.3  | 33.9  | 35.3  | 36.0  | 26.9  | 16.5  | 15.2  |
|                                                                        |             | MCWBR  | 11.4  | 14.4  | 14.4  | 17.4  | 16.9  | 13.7  | 14.1  | 15.8  | 17.4  | 15.8  | 11.5  | 11.8  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.257 | 0.264 | 0.298 | 0.346 | 0.365 | 0.388 | 0.367 | 0.368 | 0.336 | 0.299 | 0.263 | 0.252 |
|                                                                        | taud        |        | 2.335 | 2.381 | 2.289 | 2.180 | 2.272 | 2.277 | 2.386 | 2.373 | 2.454 | 2.512 | 2.406 | 2.262 |
|                                                                        | Ebn at Noon |        | 238   | 274   | 283   | 280   | 280   | 274   | 278   | 271   | 268   | 258   | 234   | 213   |
|                                                                        | Edh at Noon |        | 18    | 24    | 32    | 41    | 40    | 40    | 35    | 34    | 28    | 21    | 17    | 16    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 257   | 528   | 860   | 1318  | 1575  | 1598  | 1695  | 1455  | 1006  | 561   | 270   | 200   |
|                                                                        | RadStd      |        | 24    | 55    | 84    | 124   | 157   | 139   | 147   | 89    | 117   | 61    | 30    | 22    |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (7 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page